

Hackathon Problem Statements

Regd No: 25091A3252, 25091A3268

Problem Statement No: 51

Metro/Bus Fare Calculator with Dynamic Pricing Create a fare calculation engine using the OOP strategy design pattern to support multiple pricing models (distance-based, zone-based, peak/off-peak) for public transport systems. Apply demand data analysis to recommend dynamic peak-hour pricing that balances revenue and ridership. Use graph algorithms to compute distance-based fares along transit routes. This helps transport authorities design fairer, demand-responsive fare structures.

Regd No: 25091A32K9, 25091A32J2

Problem Statement No: 88

Microplastic Contamination Mapping Build a microplastic contamination mapping tool using spatial clustering on water/soil sample datasets to identify contamination hotspots along rivers or coastlines. Visualize concentration patterns to guide targeted cleanup and source-reduction efforts. Use OOP to model sample sites, contamination levels, and monitoring records. This raises data-driven awareness of an increasingly critical but under-monitored pollution issue.

Regd No: 25091A32H5, 25091A32M1

Problem Statement No: 117

Search-and-Rescue Drone Path Optimizer Create a drone path planning tool using graph traversal algorithms to compute optimal search patterns over a disaster-affected area, maximizing area coverage within battery/time constraints. Use OOP to model search grids, drones, and detected points-of-interest. This supports modern disaster response teams increasingly relying on drones for faster search-and-rescue operations in inaccessible terrain.

Regd No: 25091A3260, 25091A32E9

Problem Statement No: 124

Voter Turnout Prediction System Create a voter turnout prediction tool using regression analysis on demographic and historical turnout datasets to forecast expected participation rates across constituencies, helping election commissions plan polling infrastructure and awareness campaigns for low-turnout areas. Use OOP to model constituencies, historical data, and predictions. This supports more efficient allocation of voter awareness resources.

Regd No: 25091A3237, 25091A3214

Problem Statement No: 126

Elderly Pension Disbursement Tracker Design a pension tracking system using OOP-based transaction and state management to monitor disbursement status, flag delayed payments, and generate automated reminders for pending verifications. This addresses common pension delivery issues faced by elderly beneficiaries in rural areas, who often lack the means to follow up on bureaucratic delays themselves.

Regd No: 25091A3219, 25091A32A7

Problem Statement No: 136

Anganwadi Resource Allocation Optimizer Create a resource allocation tool using constraint-based optimization to distribute limited nutrition and educational resources across Anganwadi centers based on child enrollment numbers and need-severity indicators. Use OOP to model centers, resources, and allocation rules. This supports more equitable early childhood care resource distribution in a system that currently varies widely by local administrative capacity.